How AI Works in LexAnalyzer

Plain-English Guide: Document Q&A, Summaries, Clause Risk & Blind-Spot Detection

1. How AI Answers Your Questions (Document Chat)

LexAnalyzer uses **Retrieval-Augmented Generation (RAG)**. Instead of guessing, the AI answers strictly from your uploaded contract:

- **1. Vector Matching:** When you type a question, local AI models (*all-MiniLM-L6-v2*) convert your question into a mathematical search vector.
- **2. FAISS Semantic Search:** The system scans the document's local vector index and retrieves the top 4 exact paragraphs that match your question conceptually.
- **3. Strict Grounding:** The retrieved contract text is fed to the LLM with strict instructions: *'Only answer using this text. Do not make anything up.'*
- **4. Verified Citations:** The AI provides a clear plain-English answer and attaches exact page numbers so you can verify the clause in the PDF.

2. How Document Summaries are Generated

Instead of reading 50 pages of legal jargon, the AI distills the agreement into executive-ready takeaways:

- **1. Structure Parsing:** The AI scans key anchors across all pages (Recitals, Scope, Financial Terms, Term & Termination).
- **2. Commercial Extraction:** It identifies the core business deal: Who is paying whom? What are the deliverables? What are the key milestones?
- **3. Four-Pillar Summary:** The output is organized into crisp sections: *Executive Overview*, *Key Obligations*, *Liabilities & Deadlines*, and *Practical Takeaways*.
- **4. Plain-English Translation:** Archaic legal phrases (e.g., *'heretofore'*, *'indemnify and hold harmless'*) are translated into clear business language.

3. How Clause Analysis & Risk Profiling Works

The AI systematically reviews the agreement to identify legal commitments and assess their risk level:

- **1. Clause Classification:** Automatically identifies key clause types: Indemnification, Governing Law, Limitation of Liability, Termination, Confidentiality, and Dispute Resolution.
- **2. Balance Assessment:** The AI checks whether obligations are mutual or heavily one-sided favoring the other party.
- **3. Risk Grading:**
  - **HIGH RISK:** Unlimited liability, one-sided indemnities, automatic renewal with short opt-out windows.
  - **MEDIUM RISK:** Ambiguous termination triggers or unfamiliar foreign governing jurisdictions.
  - **LOW RISK:** Standard bilateral confidentiality and mutual dispute resolution terms.
- **4. Actionable Advice:** For every flagged risk, the AI explains *why* it is dangerous and suggests amendment language.

4. How the Contract Blind-Spot Detector Works

Traditional tools only review what is written. The **Blind-Spot Detector** scans for what is **MISSING, VAGUE, or UNRESOLVED**:

- **1. Potential Gaps:** Detects missing protective clauses that should be in the contract (e.g., missing cap on damages, missing audit rights, no force majeure).
- **2. Ambiguity Detection:** Highlights vague wording like *'reasonable efforts'*, *'timely manner'*, or *'to the satisfaction of'* that lead to costly disputes.
- **3. Missing References:** Flags references to unattached documents (e.g., contract says *'as defined in Exhibit B'*, but Exhibit B is missing from the upload).
- **4. Questions for Human Counsel:** Generates targeted legal questions to ask the counterparty during negotiation.

Key Takeaway: LexAnalyzer combines local vector search (FAISS) with high-speed LLM reasoning. It never relies on blind AI hallucination — every answer, risk score, and detected blind spot is anchored directly in verifiable text with page references.